

Inverse Line-Heating Plan Demo: Process, Mathematics, and Results

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1 Overview

This document records the inverse planning run for the single-line heating case (plan_demo), including the mathematical model, the optimization process, a flow chart of the workflow, and the full optimization log. The run targets a midspan deflection profile on a 400 mm $\times$ 300 mm plate of thickness 14 mm.

2 Mathematical Model

2.1 Gaussian Heat Source

The surface heat flux is modeled as a Gaussian:

$$q(r) = q_0 \exp\left(-\frac{2r^2}{r_0^2}\right), \quad (1)$$

where r_0 is the heat source radius. The total power is

$$P = \int_0^\infty 2\pi r q(r) dr = \frac{\pi r_0^2}{2} q_0. \quad (2)$$

With line energy input E_{in} and scan speed v , power is $P = E_{\text{in}}v$, giving

$$q_0 = \frac{2E_{\text{in}}v}{\pi r_0^2}. \quad (3)$$

2.2 Objective

The planner minimizes a scalar loss between the simulated deflection profile and the target profile. For the single-line case, the target is the midline deflection profile:

$$\{(y_i, w_i)\}_{i=1}^3 = \{(0, 0), (150, 5.8), (300, 0)\} \text{ mm}. \quad (4)$$

The reported objective value is the scalar loss returned by the planner for each evaluation.

3 Process

3.1 Inputs

- Plate: $L_x = 400$ mm, $L_y = 300$ mm, thickness 14 mm.
- Initial line: $y = 150$ mm, $E = 600$ J/mm, $v = 6$ mm/s, 1 pass.
- Bounds: $y \in [50, 250]$, $E \in [400, 1000]$, $v \in [4, 20]$, passes $\in [1, 3]$.
- Solver: Gaussian radius $r_0 = 7$ mm, $\beta = 3$, $\Delta t = 2$ s, extra time 200 s, centerline-fixed boundary condition, inherent strain enabled.

3.2 Workflow (Flow Chart)

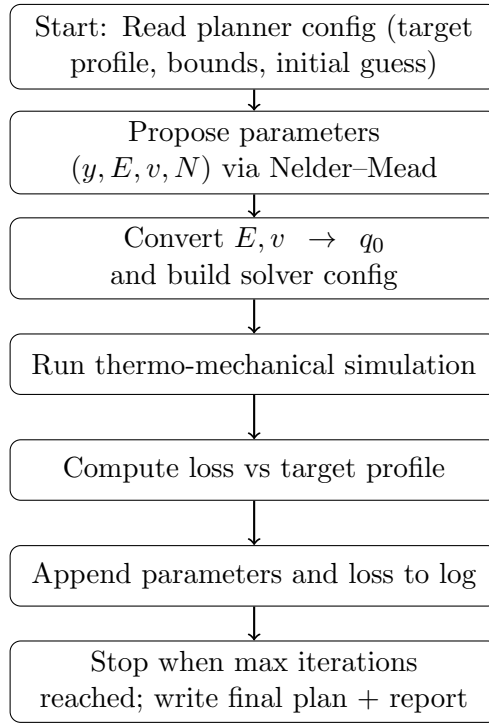


Figure 1: Inverse planning workflow.

4 Results

4.1 Final Plan

- Line position: $y = 150.4926$ mm
- Energy: $E = 602.6605$ J/mm
- Velocity: $v = 6.0631$ mm/s
- Passes: $N = 1$
- Objective: 3.886×10^{-4}

4.2 Deflection Summary

- $w_{\max} = 5.8341$ mm
- $w_{\min} = -0.5943$ mm
- Edge-to-edge camber: 0.0847 mm

5 Optimization Log (CSV)

```
"y_lines,energy,velocity,passes,loss
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150.000,600.0,6.0,1,0.2674439275059965
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"

6 References to Generated Artifacts

- Plan JSON: `results/plans/plan_demo.json`
- Simulation summary: `results/plans/plan_demo/summary.json`
- Report PDF: `results/plans/plan_demo/report.pdf`
- Optimization log: `results/plans/plan_log.csv`